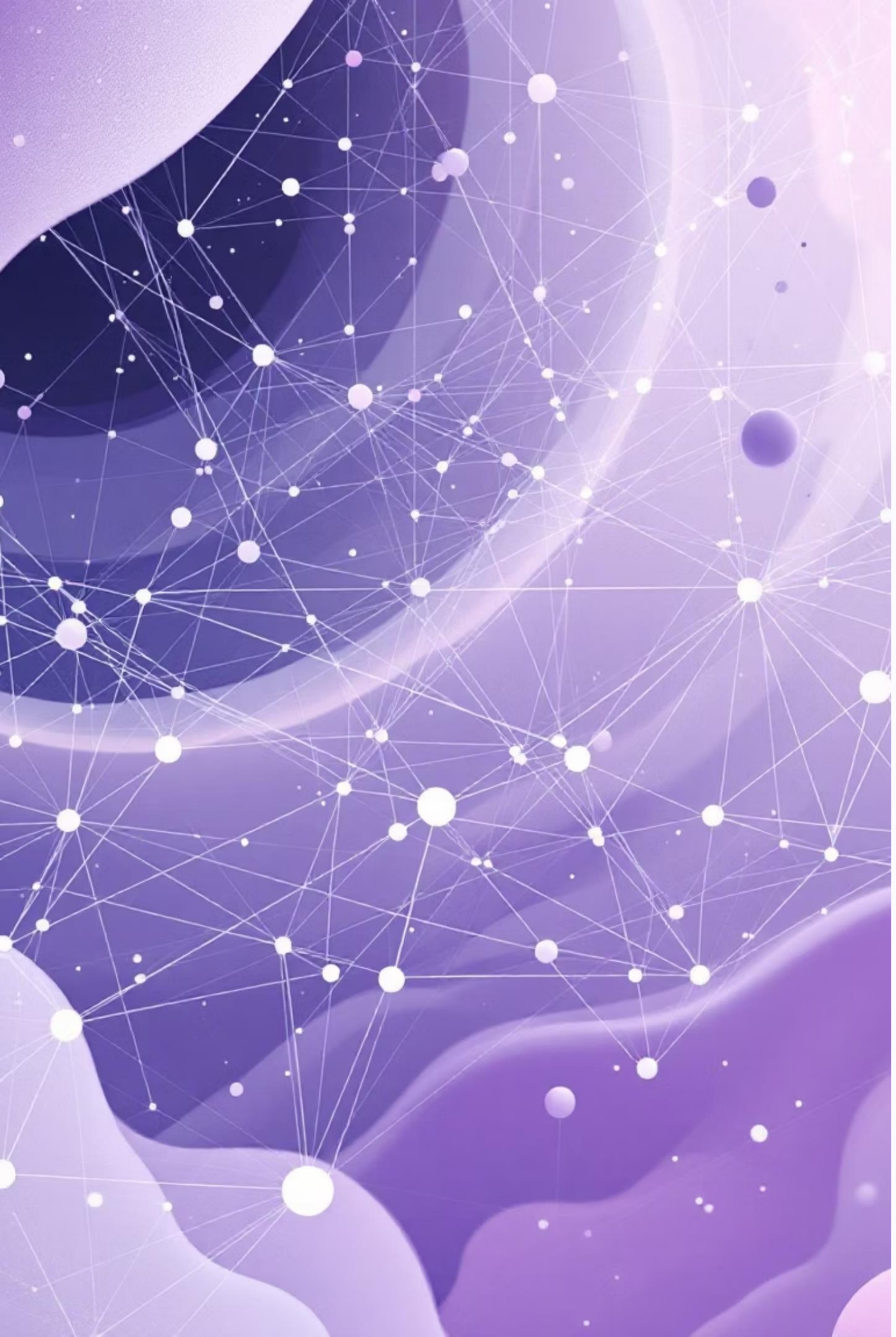

Data Visualization Project: Dissecting the world development indicators(WDI)

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Unlocking Global Insights: Key Research Questions

Top 7 Main Indicator Categories

This analysis identifies which thematic domains (such as Economy or Health) have the highest volume of indicators to determine the dataset's primary areas of focus.

Assessing the Completeness of Measurement Units

This examines the "Unit of Measure" field to distinguish between "Ready-to-Use" quantitative indicators and "Metadata-Heavy" qualitative indices.

Metadata Coverage Across Major Development Topics

This analyzes the presence of methodology, limitations, and relevance fields to evaluate the transparency and documentation quality of each topic.

Distribution of Indicators Across Main Development Topics

This utilizes the Pareto principle to discover if a small number of topics dominate the dataset and to visualize the cumulative share of indicators.

Project Overview & Dataset Selection

Our project focuses on analyzing the metadata and structure of the World Development Indicators (WDI) provided by the World Bank. This allows us to understand the scope and composition of one of the world's most critical open data repositories.

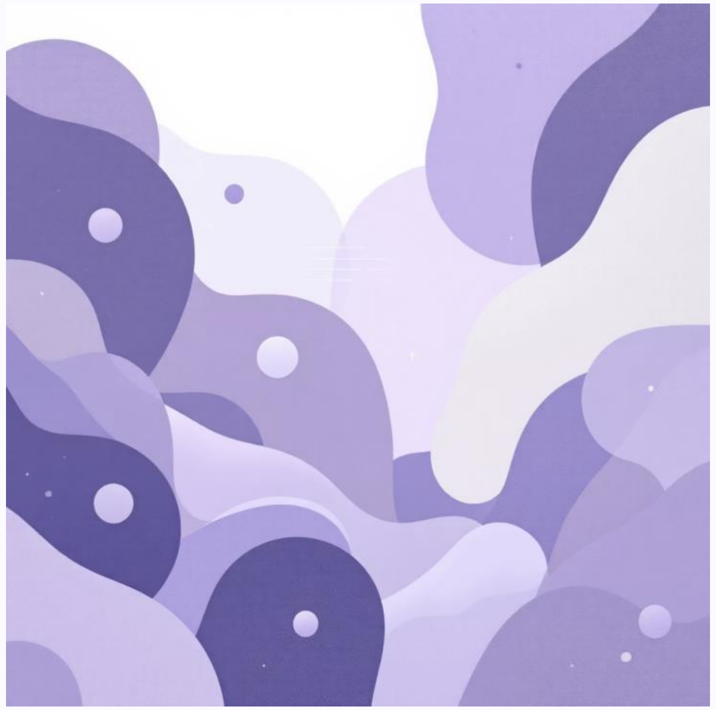
The dataset consists of metadata describing development indicators, including topic classifications, definitions, sources, methodological notes, limitations, and relevance information. Initial preprocessing involved inspecting data types, identifying missing values, and assessing overall metadata completeness across features.

To standardize topic analysis, a new variable called Main Topic was created by extracting the high-level category from the original Topic field. Missing topic values were labeled as Unknown. This step enabled consistent aggregation and comparison across thematic domains.

Methodology: From Raw Data to Insight

Tools & Libraries

- Programming Language: Python was used for all data processing and visualization tasks.
- Data Manipulation: Pandas and NumPy were utilized to clean, filter, and transform the WDI metadata.
- Visualization: The first way is Matplotlib and Seaborn were employed to create technical charts, including Donut, Bar, and Pareto plots and the second way is using excel.
- Platform: Analysis was conducted within Jupyter Notebooks for iterative development.



Data Cleaning & Preprocessing

- Standardization: Extracted a new "Main Topic" variable from the raw Topic field to allow for high-level thematic aggregation.
- Missing Value Management: Identified missing metadata and labeled empty Topic fields as "Unknown" to ensure data integrity during analysis.
- Classification: Categorized indicators into "Ready-to-Use" (with units) and "Metadata-Heavy" (qualitative/indices) based on measurement completeness.

Insights from the Data

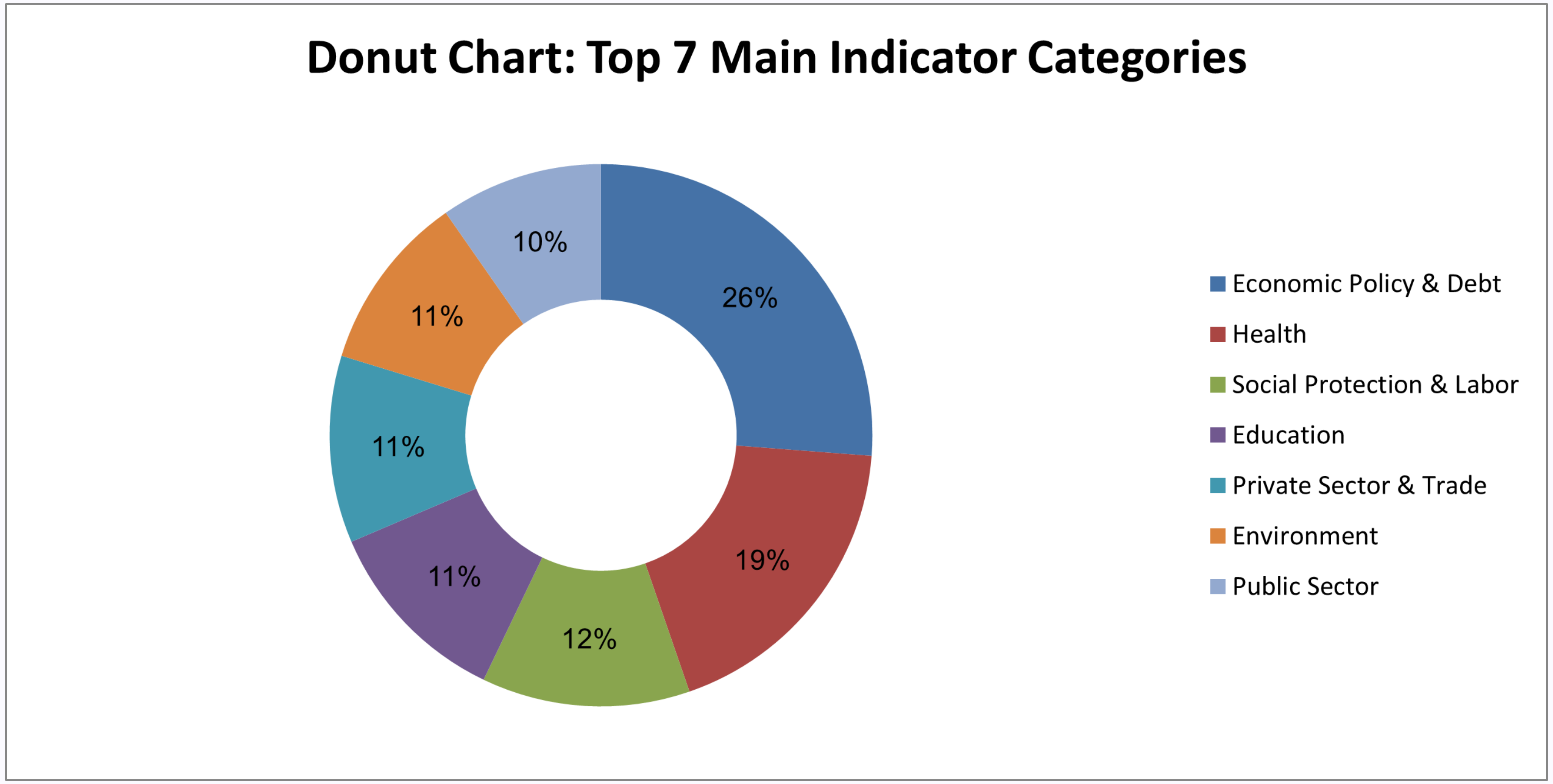
The quality and transparency of metadata are not consistent across the dataset. While most indicators provide methodological information, there are significant gaps in the reporting of limitations and exceptions in certain thematic areas..

The dataset exhibits a high degree of concentration, where a small number of core development topics contain the vast majority of the indicators.

Certain development areas, such as Gender, Infrastructure, and Trade, are represented by a much smaller volume of indicators compared to dominant fields like Social Protection and Labor.

The thematic focus is heavily skewed toward specific domains, with "Economic Policy & Debt" and "Health" together representing nearly half of the analyzed indicators.

Visualization 1: Top 7 Main Indicator Categories



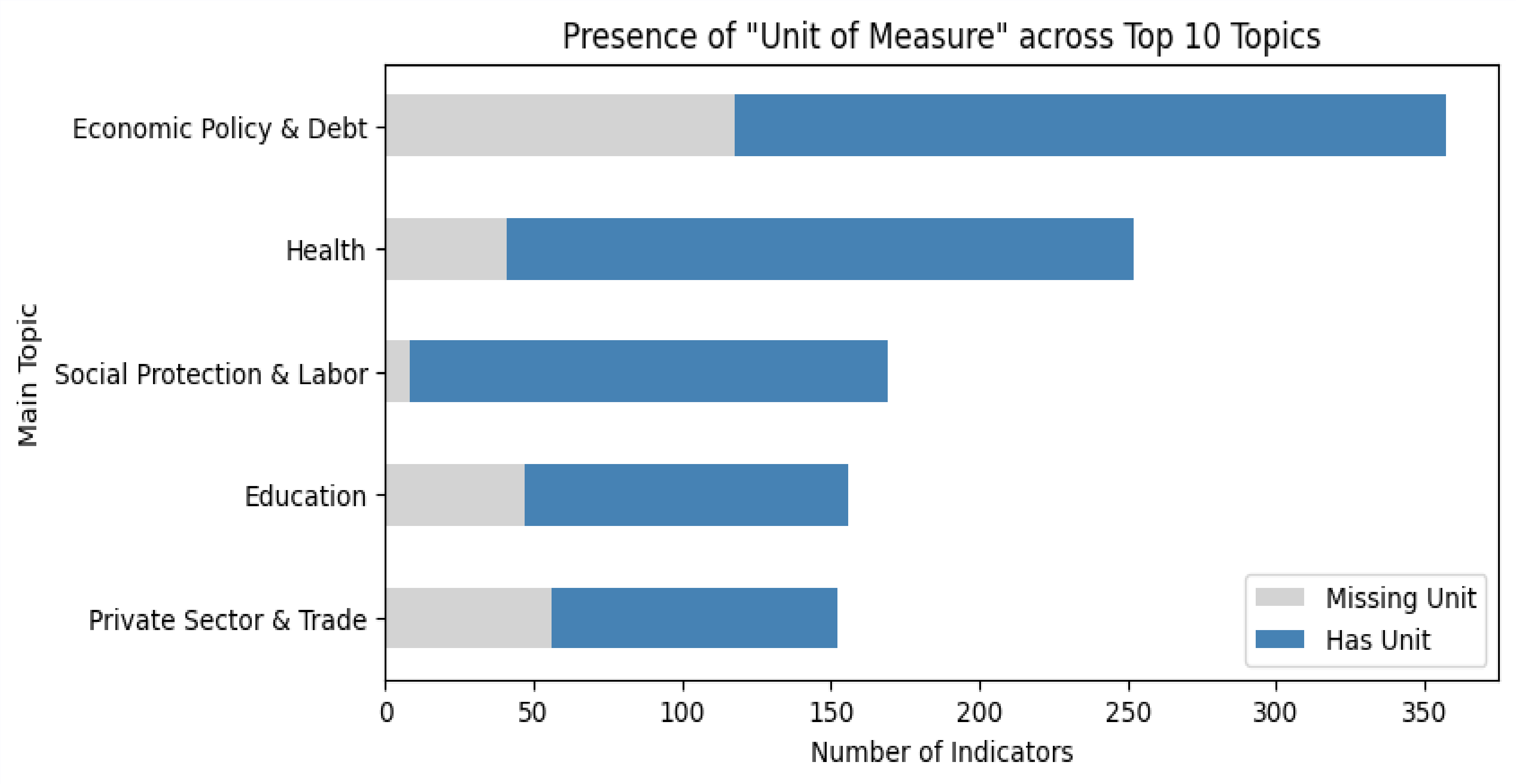
This donut chart shows the **relative distribution of indicators** among the seven most frequent main topic categories in the dataset.

Each segment represents the **percentage share of indicators** belonging to a specific main topic, allowing for an easy comparison of their relative importance.

The visualization highlights that a small number of topics account for a large proportion of indicators, while the remaining topics have a smaller share.

This indicates an **uneven distribution of indicators across topics**, with certain domains being more heavily represented in the dataset.

Visualization 2: Assessing the completeness of measurement units in the Top 5 Categories



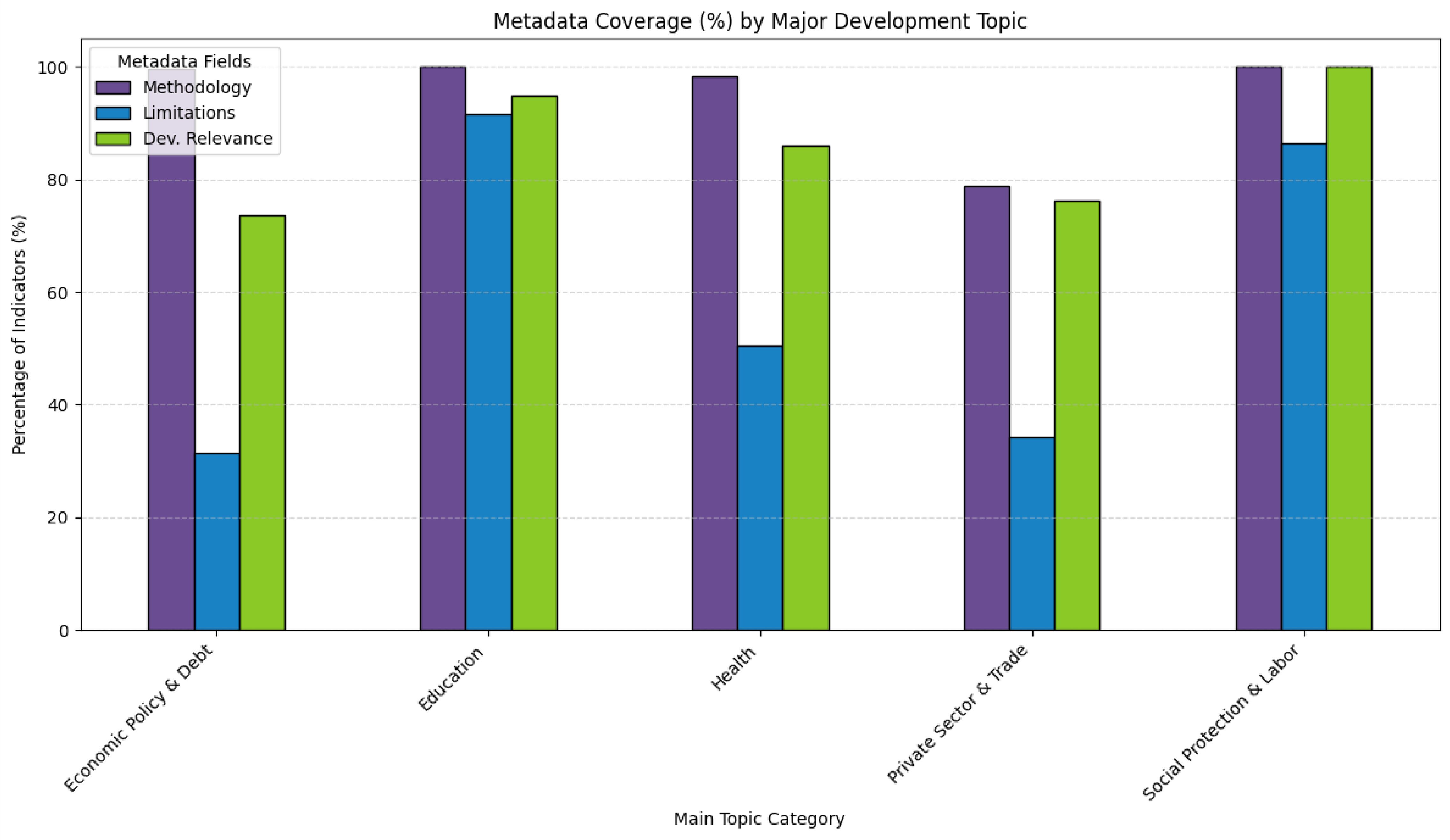
This plot answers the question, "Do we actually know how to measure these indicators?"

Steel Blue (Has Unit): These are "Ready-to-Use" indicators. They have clear units like "Percentage of GDP" or "Years," making them easy to plot in a chart immediately.

Light Grey (Missing Unit): These are "Metadata-Heavy" indicators. They are often indices, scores, or qualitative measures that don't have a single physical unit (like an "Ease of Doing Business" score).

Comparison: You can see that **Economic Policy & Debt** has the most "Missing Units." This makes sense because many economic metrics are complex ratios or scores rather than simple counts.

Visualization 3:Metadata Coverage Across Major Development Topics

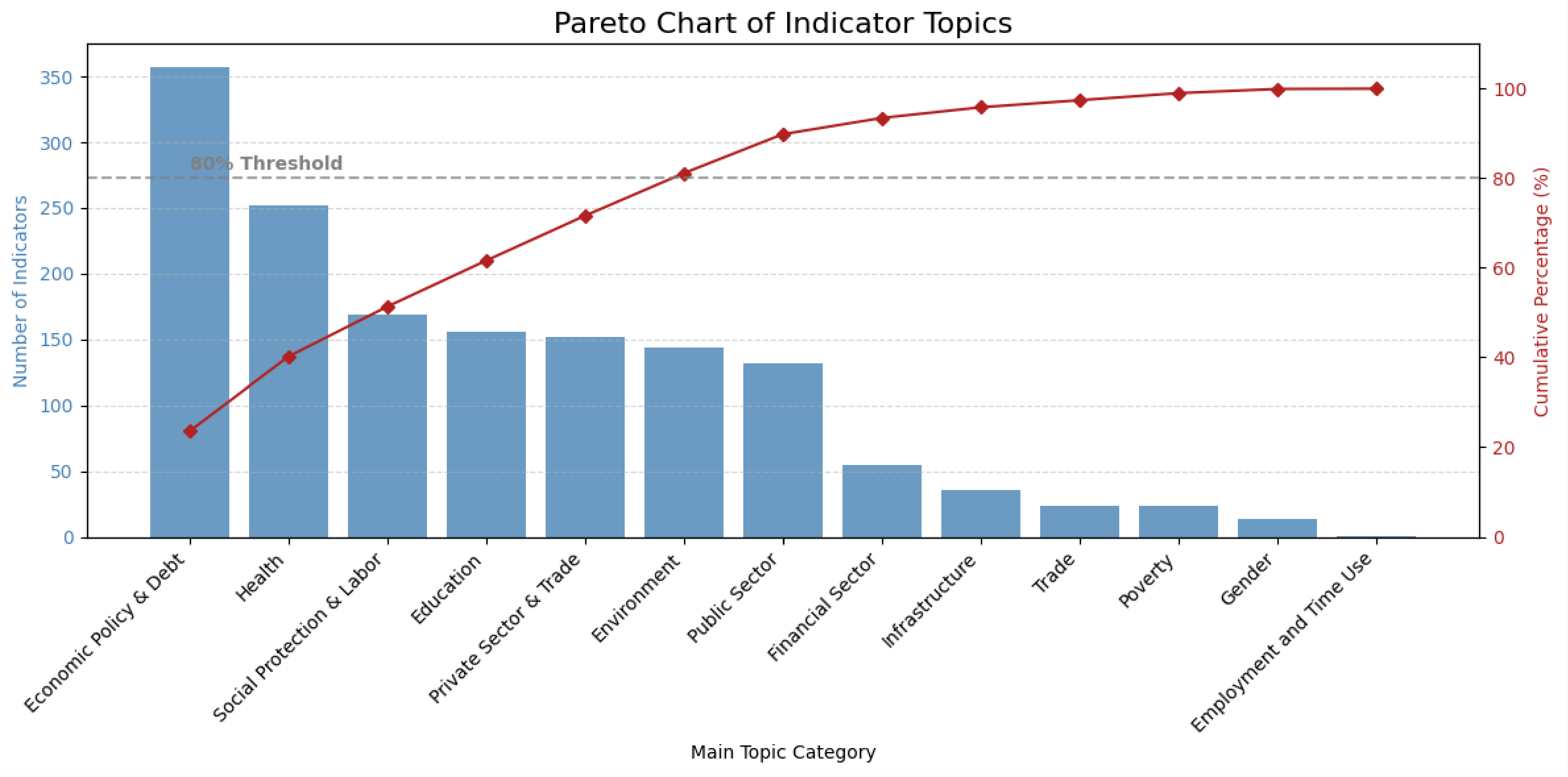


This chart shows how well key metadata fields are documented across the five most common development topics in the dataset.
For each topic, the bars represent the percentage of indicators that include information on statistical methodology, limitations and exceptions, and development relevance

The results highlight noticeable differences in documentation quality between topics.
While some areas show strong coverage—indicating well-described and transparent indicators—others have gaps, particularly in the reporting of limitations or methodological details.

Overall, the chart emphasizes that metadata completeness is not uniform across topics and identifies areas where improved documentation would enhance data usability, comparability, and reliability.

Visualization 4: Distribution of Indicators Across Main Development Topics



This chart illustrates how indicators are distributed across main topic categories.
The **blue bars** represent the number of indicators in each main topic, ordered from most to least frequent.
The **red line** shows the cumulative percentage of indicators as topics are added from left to right.

The chart helps identify which topics account for the majority of indicators in the dataset.
The dashed **80% reference line** highlights the point at which a small number of topics collectively contribute to most of the indicators, reflecting the Pareto principle (80/20 rule).

This visualization shows that a limited set of main topics dominates the dataset, while the remaining topics contribute relatively fewer indicators.

Challenges & Lessons Learned

Data Cleaning Complexities

Managing missing values and standardizing diverse data formats proved challenging, emphasizing the importance of robust pre-processing.

Design for Clarity

Balancing data density with visual simplicity was key. Iterating on color palettes and annotation strategies significantly improved readability.

Narrative-Driven Visualization

Focusing on the core research question helped refine plot choices and ensured each visualization contributed meaningfully to the story.

Unexpected Discoveries

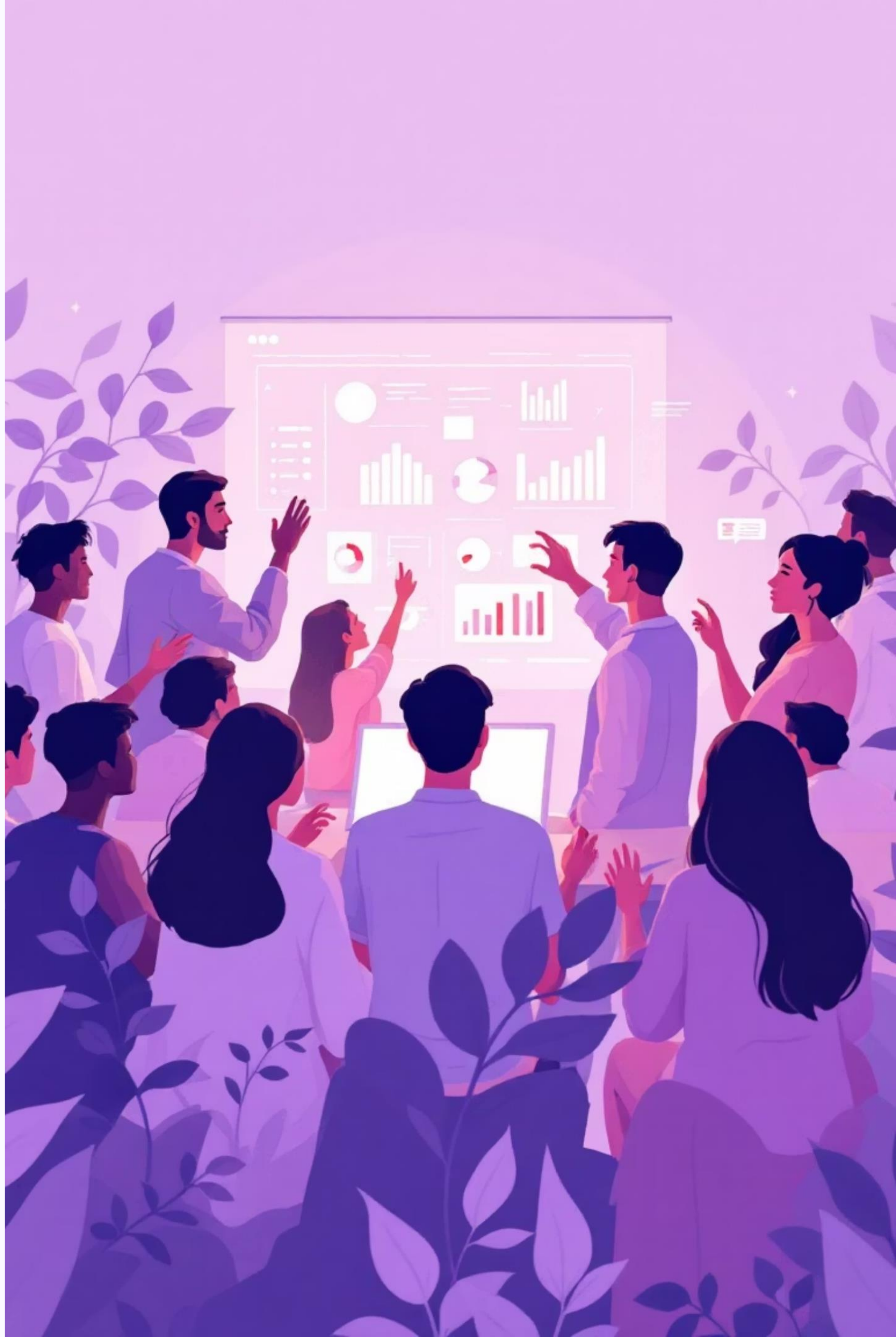
The data revealed a surprisingly strong correlation between certain policy interventions and renewable adoption rates, prompting deeper dives.

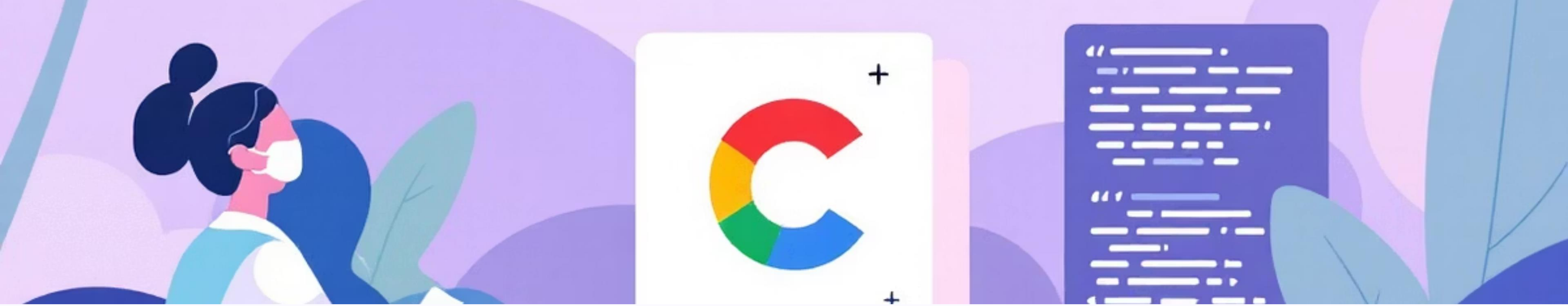
Key Insights

1. Indicator coverage is highly concentrated: a small number of main topics account for the majority of indicators, as shown by both the Pareto and donut charts.

2. Topic distribution is uneven, with several topics being heavily represented while many others contribute only a small share of indicators

3. Metadata completeness varies by topic: some major topics consistently include methodology, limitations, and development relevance, while others show noticeable documentation gaps.





Data & Code Availability

Transparency and reproducibility are fundamental to scientific inquiry. All data used and code developed for this analysis are openly accessible for review and further exploration.

Access Code on GitHub

Explore On Kaggle

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